FEM - Solution to Mechanical Equilibrium and Special Elements Sections

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Contents

Introduction

This document seeks to organize the solutions to the chapters 4 and 5 of Finite Elements solutions course at *Universidade de Brasília*.

1. Theorical Solutions

1.1 Exercise 4.1.1

In which cases the strain tensor is symmetric?

The strain tensor is symmetric in all cases where the small displacement assumption holds, as it is defined as the symmetric part of the displacement gradient.

1.2 Exercise 4.1.2

Why is the stress tensor always symmetric?

asdf

1.3 Exercise 4.1.3

Why does the classical finite element formulation assume small deformations?

asdf

1.4 Exercise 4.1.4

Give examples of typical structures corresponding to each stress state: plane stress, plane strain, and axisymmetric conditions. Provide at least three examples for each case.

asdf

1.5 Exercise 4.2.1

Explain the role of the strain-displacement matrix $\boldsymbol{B}$ in a finite element formulation.

asdf

1.6 Exercise 4.3.1

What is an equivalent nodal-load vector, and why is it needed in the finite element method?

asdf

1.7 Exercise 4.3.2

Explain the difference between an externally applied nodal force and a consistent equivalent nodal load obtained from a distributed action.

asdf

1.8 Exercise 4.3.3

How do body forces and surface tractions enter the weak form and the element load vector?

asdf

1.9 Exercise 4.4.1

What is full and reduced integration in the computation of the stiffness matrix?

asdf

1.10 Exercise 4.4.2

What are the connectivity and location vectors of an element, and how are they used in the assembly process?

asdf

1.11 Exercise 4.4.3

What is the equilibrium residual, and why must it vanish only at free degrees of freedom?

asdf

1.12 Exercise 4.4.4

How are support reactions recovered after the displacement field has been computed?

asdf

1.13 Exercise 4.5.1

Explain the origin of shear locking and volumetric locking, and indicate in which types of problems each one is most likely to appear.

asdf

1.14 Exercise 4.5.2

Why can reduced integration alleviate locking, and what numerical risk does it introduce?

asdf

1.15 Exercise 4.5.3

What are hourglass modes, and why are they associated with underintegrated elements?

asdf

1.16 Exercise 4.5.4

Why can element distortion degrade accuracy even when the interpolation order is unchanged?

asdf

1.17 Exercise 4.5.5

What is meant by ill-conditioning of the stiffness matrix, and how can it affect the numerical solution?

asdf

1.18 Exercise 4.5.6

Distinguish between an ill-conditioned stiffness matrix and a singular stiffness matrix.

asdf

1.19 Exercise 4.5.7

Give four common modeling or formulation errors that can lead to a singular stiffness matrix.

asdf

1.20 Exercise 4.5.8

Why may excessively large penalty parameters create numerical difficulties even though they enforce constraints more strongly?

asdf

1.21 Exercise 4.6.1

List five use-cases of multifreedom constraints in finite-element modeling.

asdf

1.22 Exercise 4.6.2

Compare master-slave elimination, the penalty method, and the Lagrange multiplier method for enforcing constraints. State one advantage and one drawback of each.

asdf

1.23 Exercise 5.1.1

List four applications for each of the following element types: plate, membrane, shell, and interface.

asdf

1.24 Exercise 5.1.3

Simplify the 3D formulation for isoparametric beam elements in order to derive the formulation for an isoparametric 2D beam element.

asdf

1.25 Exercise 5.1.4

Derive the formulation for a 2D interface element.

asdf

2. Exercise 4.1.5

A flat metallic plate with length 100 mm, width 40 mm, and thickness 5 mm is subjected to a uniform tensile load. When the axial force was measured as 20 kN the axial elongation was 0.1 mm and in-plane lateral contraction 0.008 mm. Determine the corresponding constitutive matrix **D** for a plane stress 2D linear elastic model.

2.1 Solution

Firstly, we need to calculate the tensions

$$\sigma_x = \frac{F}{A} = \frac{20000\ \mathrm{N}}{0.0002\mathrm{m}^2} = 100\ \mathrm{MPa} \tag{4.1.5.1}$$

$$\varepsilon_x = \frac{\Delta L}{L} = \frac{0.0001\ \mathrm{m}}{0.1\ \mathrm{m}} = 0.001\ \mathrm{m/m} \tag{4.1.5.2}$$

It is known that $E = \frac{\sigma}{\varepsilon}$, so:

$$E = \frac{100000000}{0.001} = 100000000000\ \mathrm{Pa} = 100\ \mathrm{GPa} \tag{4.1.5.3}$$

Similarly

$$\varepsilon_y = \frac{\Delta W}{W} = \frac{-0.000008}{0.04} = -0.0002 \tag{4.1.5.4}$$

$$\nu = -\frac{\varepsilon_y}{\varepsilon_x} = \frac{-0.0002}{0.001} = 0.2 \tag{4.1.5.5}$$

The constitutive Matrix is then defined by

$$\boldsymbol{D} = \frac{E}{1-\nu^2} \cdot \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \tag{4.1.5.6}$$

so

$$\boldsymbol{D} = \frac{100000000000}{1-(0.2)^2} \cdot \begin{bmatrix} 1 & 0.2 & 0 \\ 0.2 & 1 & 0 \\ 0 & 0 & \frac{1-(0.2)}{2} \end{bmatrix} \tag{4.1.5.7}$$

$$\boldsymbol{D} = 104.1667\mathrm{GPa} \cdot \begin{bmatrix} 1 & 0.2 & 0 \\ 0.2 & 1 & 0 \\ 0 & 0 & 0.4 \end{bmatrix} \tag{4.1.5.8}$$

$$\boldsymbol{D} = \begin{bmatrix} 104.1667 & 20.8333 & 0 \\ 20.8333 & 104.1667 & 0 \\ 0 & 0 & 41.6667 \end{bmatrix} \mathrm{GPa} \tag{4.1.5.9}$$

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3. Exercise 4.1.6

A cylindrical concrete sample with height 20 cm and diameter 10 cm is tested in uniaxial compression. At a point within the elastic range, the measured values are: axial stress 20 MPa, axial shortening 0.2 mm, and radial expansion 0.015 mm. Using these measurements, determine the elastic parameters E and ν, and construct the corresponding constitutive matrix ***D*** for a plane stress linear[1] elastic model.

3.1 Solution

Firstly, we need to define Young's module:

$$E = \frac{\sigma}{\varepsilon} \tag{4.1.6.1}$$

$$\varepsilon = \frac{\Delta L}{L} \tag{4.1.6.2}$$

Substituting (4.1.6.2) in (4.1.6.1), we get

$$E = \frac{\sigma \cdot L}{\Delta L} = \frac{20 \text{ MPa} \cdot 20 \text{ cm}}{0.02 \text{ cm}} = 20 \text{ GPa} \tag{4.1.6.3}$$

Now, we may obtain epsilon in relation to the radius:

$$\varepsilon_z = \frac{\Delta L}{L} = -0.001 \text{ m/m} \tag{4.1.6.4}$$

$$\varepsilon_r = \frac{\Delta r}{r} = 0.0003 \text{ m/m} \tag{4.1.6.5}$$

Defining Poisson:

$$\nu = -\frac{\varepsilon_r}{\varepsilon_z} = 0.3 \tag{4.1.6.6}$$

Firstly, let's analyse the constitutive matrix for an axissimetrical element, defined as

$$\begin{bmatrix} \sigma_{\text{rr}} \\ \sigma_{\text{zz}} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{bmatrix} = \boldsymbol{D} \cdot \begin{bmatrix} \varepsilon_{\text{rr}} \\ \varepsilon_{\text{zz}} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{bmatrix} \tag{4.1.6.7}$$

Where **D** is defined by

$$\boldsymbol{D} = \frac{E}{(1+\nu)(1-2\nu)} \cdot \begin{bmatrix} 1-\nu & \nu & \nu & 0 \\ \nu & 1-\nu & \nu & 0 \\ \nu & \nu & 1-\nu & 0 \\ 0 & 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \tag{4.1.6.8}$$

$$\boldsymbol{D} = 38461.5385 \text{ MPa} \cdot \begin{bmatrix} 0.7 & 0.3 & 0.3 & 0 \\ 0.3 & 0.7 & 0.3 & 0 \\ 0.3 & 0.3 & 0.7 & 0 \\ 0 & 0 & 0 & 0.2 \end{bmatrix} \tag{4.1.6.9}$$

[1] I've read the problem incorrectly and done also the axissimetrical development, oops!

$$= \begin{bmatrix} 26923.1 & 11538.5 & 11538.5 & 0 \\ 11538.5 & 26923.1 & 11538.5 & 0 \\ 11538.5 & 11538.5 & 26923.1 & 0 \\ 0 & 0 & 0 & 7692.3 \end{bmatrix} \text{MPa} \tag{4.1.6.10}$$

Finally, to define the same element in a plane stress linear elastic model:

$$\boldsymbol{D} = \frac{E}{1-\nu^2} \begin{pmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{pmatrix} \tag{4.1.6.11}$$

$$\boldsymbol{D} = \frac{20000 \text{ MPa}}{1-0.3^2} \cdot \begin{bmatrix} 1 & 0.3 & 0 \\ 0.3 & 1 & 0 \\ 0 & 0 & 0.35 \end{bmatrix} \tag{4.1.6.12}$$

$$\boldsymbol{D} = \begin{bmatrix} 2.1978 & 0.6593 & 0 \\ 0.6593 & 2.1978 & 0 \\ 0 & 0 & 0.7692 \end{bmatrix} \cdot 10^4 \text{ MPa} \tag{4.1.6.13}$$

4. Exercise 4.1.7

A long thick-walled cylindrical pipe with inner radius a = 50 mm and outer radius b = 100 mm is subjected to a uniform internal pressure p = 10 MPa. The material is linear elastic and isotropic, with Young's modulus E = 200 GPa and Poisson's ratio ν = 0.2. Assume that the pipe is sufficiently long so that the axial strain can be neglected, i.e. εz = 0. At a point located at r = 75 mm within the wall, the stress state is characterized by normal stresses only, with no shear components. Determine the stress components at this point and the corresponding strain components.

4.1 Solution

Since the internal pressure acts in the radial direction, and assuming axial strength can be neglected, it is valid to assume that the problem constitutes an Axisymmetric Stress, defined by

$$\boldsymbol{\sigma} = \begin{bmatrix} \sigma_{rr} & 0 & \sigma_{rz} \\ 0 & \sigma_{\theta\theta} & 0 \\ \sigma_{zr} & 0 & \sigma_{zz} \end{bmatrix} \tag{4.1.7.1}$$

It is known, for the current material, that

$$\nu = 0.2 \tag{4.1.7.2}$$

$$E = 200 \text{ MPa} \tag{4.1.7.3}$$

Thus, for the Axisymmetric condition:

$$\begin{bmatrix} \sigma_{\mathrm{rr}} \\ \sigma_{\mathrm{zz}} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{bmatrix} = \boldsymbol{D} \cdot \begin{bmatrix} \varepsilon_{\mathrm{rr}} \\ \varepsilon_{\mathrm{zz}} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{bmatrix} \tag{4.1.7.4}$$

But, since axial strain can be neglected, that is, $\varepsilon_{zz} = 0$, we can define it as

$$\begin{bmatrix} \sigma_{\mathrm{rr}} \\ \sigma_{\theta\theta} \\ \tau_{rz} \end{bmatrix} = \boldsymbol{D} \cdot \begin{bmatrix} \varepsilon_{\mathrm{rr}} \\ \varepsilon_{\theta\theta} \\ \gamma_{rz} \end{bmatrix} \tag{4.1.7.5}$$

Where $\boldsymbol{D}$, in this specific case, will be

$$\boldsymbol{D} = \frac{E}{(1+\nu)(1-2\nu)} \cdot \begin{bmatrix} 1-\nu & \nu & 0 \\ \nu & 1-\nu & 0 \\ 0 & 0 & \frac{1-2\nu}{2} \end{bmatrix} \tag{4.1.7.6}$$

$$\boldsymbol{D} = \frac{200}{(1+0.2)(1-0.4)} \cdot \begin{bmatrix} 0.8 & 0.2 & 0 \\ 0.2 & 0.8 & 0 \\ 0 & 0 & 0.3 \end{bmatrix} \tag{4.1.7.7}$$

$$\boldsymbol{D} = \begin{bmatrix} 222.2222 & 55.5556 & 0 \\ 55.5556 & 222.2222 & 0 \\ 0 & 0 & 83.3333 \end{bmatrix} \text{MPa} \tag{4.1.7.8}$$

With the defined matrix:[2]

[2]This problem is incomplete

5. Exercise 4.2.2

Build the matrix B for a three-node bar element with nodal coordinates (x_1, y_1), (x_2, y_2), and (x_3, y_3).

5.1 Solution

We can estimate the displacement with the use of shape functions:

$$\boldsymbol{u}(x, y, z) \approx \boldsymbol{N}(x, y, z)\boldsymbol{U}^{(e)} \tag{4.2.2.1}$$

Where $\boldsymbol{U}^{(e)}$ is a collection of the displacement degrees of freedom.

Analysing a single node in its x displacement, it is true that

$$u_{x(x,y,z)} = \sum_{i=1}^{n} N_{i(x,y,z)} u_{ix} \tag{4.2.2.2}$$

Deriving in relation to x:

$$\varepsilon_{xx} = \frac{\partial u_x}{\partial x} = \sum_{i=1}^{n} \frac{\partial N_i}{\partial x} u_{ix} \tag{4.2.2.3}$$

Thus, the $\boldsymbol{B}$ matrix represents an element to generalize the relation between the strain vector and displacement matrix:

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \boldsymbol{B} \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ \vdots \\ u_{nx} \\ u_{ny} \end{bmatrix} \tag{4.2.2.4}$$

which is simply $\boldsymbol{\varepsilon} = \boldsymbol{B}\boldsymbol{U}^{(e)}$. $\boldsymbol{B}$ on itself can be represented as a union of matrixes:

$$\boldsymbol{B} = (\boldsymbol{B}_1 \;\; \boldsymbol{B}_2 \;\; \boldsymbol{B}_3 \;\; \boldsymbol{B}_4 \;\; \boldsymbol{B}_5 \;\; \dots \;\; \boldsymbol{B}_n) \tag{4.2.2.5}$$

Since our element can only suffer strain from compression/traction - that is, we assume it to be that of a truss, our equilibrium will be

$$\varepsilon_{ss} = \boldsymbol{B}\boldsymbol{U}^{(e)} = [\boldsymbol{B}_1 \;\; \boldsymbol{B}_2 \;\; \boldsymbol{B}_3] \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{bmatrix} \tag{4.2.2.6}$$

Where

$$\boldsymbol{B}_i = \begin{pmatrix} \frac{\partial N_i}{\partial x} & \frac{\partial N_i}{\partial y} \end{pmatrix} \tag{4.2.2.7}$$

To define $\boldsymbol{B}_i$, we can start from the strain definition

$$\varepsilon_{ss} = \frac{\partial u_s}{\partial s} \tag{4.2.2.8}$$

We know that, via the chain rule

$$\frac{\partial N_i}{\partial s} = \frac{dN_i}{d\xi} \cdot \frac{d\xi}{ds} \tag{4.2.2.9}$$

$$J = \|\boldsymbol{J}\| = \|\begin{pmatrix} \frac{dx}{d\xi} & \frac{dy}{d\xi} \end{pmatrix}^T\| = \sqrt{\left(\frac{dx}{d\xi}\right)^2 + \left(\frac{dy}{d\xi}\right)^2} = \frac{ds}{d\xi} \tag{4.2.2.10}$$

Hence

$$\frac{1}{J} = \frac{d\xi}{ds} \tag{4.2.2.11}$$

Substituting (4.2.2.11) into (4.2.2.9) we get

$$\frac{\partial N_i}{\partial s} = \frac{1}{J}\frac{dN_i}{d\xi} \tag{4.2.2.12}$$

$$\boldsymbol{B}_i = \begin{pmatrix} \frac{\partial N_i}{\partial x} & \frac{\partial N_i}{\partial y} \end{pmatrix} = \begin{pmatrix} \frac{\partial N_i}{\partial s}\frac{\partial x}{\partial \xi}\frac{\partial \xi}{\partial s} & \frac{\partial N_i}{\partial s}\frac{\partial y}{\partial \xi}\frac{\partial \xi}{\partial s} \end{pmatrix} = \frac{\partial N_i}{\partial s}\frac{\partial \xi}{\partial s}\begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \end{pmatrix} \tag{4.2.2.13}$$

Which finally equates to

$$\boldsymbol{B}_i = \frac{1}{J^2}\frac{dN_i}{d\xi}\begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial y}{\partial \xi} \end{pmatrix} \tag{4.2.2.14}$$

6. Exercise 4.2.3

Find the matrix $\boldsymbol{B}$ for a three-node triangular element with nodal coordinates (x_1, y_1), (x_2, y_2), and (x_3, y_3). Why is the matrix $\boldsymbol{B}$ constant along the element, and what does that imply?

6.1 Solution

First of all, a 3 node triangular element is defined by the following shape functions

$$\boldsymbol{N} = \begin{pmatrix} 1-\xi-\eta \\ \xi \\ \eta \end{pmatrix} \tag{4.2.3.1}$$

For

$$\boldsymbol{x} = \sum_{i=1}^{3} N_i(\xi, \eta) \cdot \boldsymbol{x}_i \tag{4.2.3.2}$$

$$\boldsymbol{u} = \sum_{i=1}^{3} N_i(\xi, \eta) \cdot \boldsymbol{u}_i \tag{4.2.3.3}$$

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = \boldsymbol{B} \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \end{bmatrix} \tag{4.2.3.4}$$

$$\boldsymbol{B} = (\boldsymbol{B}_1 \;\; \boldsymbol{B}_2 \;\; \boldsymbol{B}_3) \tag{4.2.3.5}$$

$$\boldsymbol{B}_i = \begin{pmatrix} \frac{\partial N_i}{\partial x} & 0 \\ 0 & \frac{\partial N_i}{\partial y} \\ \frac{\partial N_i}{\partial y} & \frac{\partial N_i}{\partial x} \end{pmatrix} \tag{4.2.3.6}$$

It is known that

$$\frac{\partial \boldsymbol{N}_i}{\partial \boldsymbol{x}} = \frac{\partial \boldsymbol{N}_i}{\partial \boldsymbol{\xi}} \boldsymbol{J}^{-1} \tag{4.2.3.7}$$

$$x(\xi, \eta) = (1-\xi-\eta)x_1 + \xi x_2 + \eta x_3 = x_1 + \xi(x_2 - x_1) + \eta(x_3 - x_1) \tag{4.2.3.8}$$

$$y(\xi, \eta) = (1-\xi-\eta)y_1 + \xi y_2 + \eta y_3 = y_1 + \xi(y_2 - y_1) + \eta(y_3 - y_1) \tag{4.2.3.9}$$

Thus the Jacobian is

$$\boldsymbol{J} = \begin{pmatrix} x_2 - x_1 & y_2 - y_1 \\ x_3 - x_1 & y_3 - y_1 \end{pmatrix} = \begin{pmatrix} x_{21} & y_{21} \\ x_{31} & y_{31} \end{pmatrix} \tag{4.2.3.10}$$

$$J = \det(\boldsymbol{J}) = x_{21}y_{31} - x_{31}y_{21} = (x_2 - x_1)(y_3 - y_1) - (x_3 - x_1)(y_3 - y_1) \tag{4.2.3.11}$$

Which, as seen by the shoelace formula, $J = 2A$.

It is known that

$$\boldsymbol{J}^{-1} = \frac{\mathrm{adj}(\boldsymbol{J})}{\det(\boldsymbol{J})} = \frac{1}{2A}\begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix} \tag{4.2.3.12}$$

For the first node

$$\begin{pmatrix} \frac{\partial N_1}{\partial x} \\ \frac{\partial N_1}{\partial y} \end{pmatrix} = \boldsymbol{J}^{-1}\begin{pmatrix} \frac{\partial N_1}{\partial \xi} \\ \frac{\partial N_1}{\partial \eta} \end{pmatrix} = \frac{1}{2A}\begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix}\begin{pmatrix} -1 \\ -1 \end{pmatrix} \tag{4.2.3.13}$$

$$\begin{pmatrix} \frac{\partial N_1}{\partial x} \\ \frac{\partial N_1}{\partial y} \end{pmatrix} = \frac{1}{2A}\begin{pmatrix} y_{21} - y_{31} \\ x_{31} - x_{21} \end{pmatrix} = \frac{1}{2A}\begin{pmatrix} y_{23} \\ x_{32} \end{pmatrix} \tag{4.2.3.14}$$

For the second node

$$\begin{pmatrix} \frac{\partial N_2}{\partial x} \\ \frac{\partial N_2}{\partial y} \end{pmatrix} = \boldsymbol{J}^{-1}\begin{pmatrix} \frac{\partial N_2}{\partial \xi} \\ \frac{\partial N_2}{\partial \eta} \end{pmatrix} = \frac{1}{2A}\begin{pmatrix} y_{31} & -y_{21} \\ -x_{31} & x_{21} \end{pmatrix}\begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{4.2.3.15}$$

$$\begin{pmatrix} \frac{\partial N_2}{\partial x} \\ \frac{\partial N_2}{\partial y} \end{pmatrix} = \frac{1}{2A}\begin{pmatrix} y_{31} \\ x_{13} \end{pmatrix} \tag{4.2.3.16}$$

For the third node

$$\begin{pmatrix} \frac{\partial N_3}{\partial x} \\ \frac{\partial N_3}{\partial y} \end{pmatrix} = \frac{1}{2A}\begin{pmatrix} y_{12} \\ x_{21} \end{pmatrix} \tag{4.2.3.17}$$

Thus, finally:

$$\boldsymbol{B} = \frac{1}{2}A \cdot \begin{pmatrix} y_{23} & 0 & y_{31} & 0 & y_{12} & 0 \\ 0 & x_{32} & 0 & x_{13} & 0 & x_{21} \\ x_{32} & y_{23} & x_{13} & y_{31} & x_{21} & y_{12} \end{pmatrix} \tag{4.2.3.18}$$

Note the notation $x_{ij} = x_i - x_j$.

The strain-displacement matrix $\boldsymbol{B}$ is uniform across the entire domain of the element Ω_e:

$$\boldsymbol{B}(x, y) = \text{const} \quad \forall (x, y) \in \Omega_e \tag{4.2.3.19}$$

This characteristic is true because the Jacobian J is constant, because the element is linear.

7. Exercise 4.2.4

For the quadrilateral element shown below, and given its nodal coordinates and nodal-displacement vector

$$C = \begin{pmatrix} 0 & 0 \\ 4 & 2 \\ 4 & 4 \\ 0 & 2 \end{pmatrix} \text{ m} \quad \text{and} \quad U^{(e)} = \begin{pmatrix} 0 \\ 0 \\ 0.01 \\ 0.01 \\ 0.015 \\ 0.015 \\ 0 \\ 0.015 \end{pmatrix} \text{ m} \tag{4.2.4.1}$$

as shown in the figure. Approximate the strain vector at the natural coordinates $(\xi, \eta) = \left(-\sqrt{\frac{1}{3}}, \sqrt{\frac{1}{3}}\right)$.

7.1 Solution

First step, is to define the shape functions and its Jacobian

$$\boldsymbol{N}(\xi, \eta) = \begin{pmatrix} \frac{(1-\xi)(1-\eta)}{4} \\ \frac{(1+\xi)(1-\eta)}{4} \\ \frac{(1+\xi)(1+\eta)}{4} \\ \frac{(1-\xi)(1+\eta)}{4} \end{pmatrix} \tag{4.2.4.2}$$

Or simply

$$N_i(\xi, \eta) = \frac{1}{4}(\xi_i - \xi)(\eta_i - \eta) \tag{4.2.4.3}$$

Taking its derivatives

$$\begin{pmatrix} \frac{\partial N_i}{\partial \xi} \\ \frac{\partial N_i}{\partial \eta} \end{pmatrix} = \begin{pmatrix} \eta - \eta_i \\ \xi - \xi_i \end{pmatrix} \tag{4.2.4.4}$$

$$\begin{pmatrix} x(\xi, \eta) \\ y(\xi, \eta) \end{pmatrix} = \sum_{i=1}^{n} \begin{pmatrix} \frac{x_i}{4}(\xi_i - \xi)(\eta_i - \eta) \\ \frac{y_i}{4}(\xi_i - \xi)(\eta_i - \eta) \end{pmatrix} \tag{4.2.4.5}$$

The strain vector for a two dimensional element, which can suffer strains in xx and yy, and the shear strain xy, is defined by

$$\begin{bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ \gamma_{xy} \end{bmatrix} = (\boldsymbol{B}_1 \;\; \boldsymbol{B}_2 \;\; \boldsymbol{B}_3 \;\; \boldsymbol{B}_4) \cdot \begin{bmatrix} u_{1x} \\ u_{1y} \\ u_{2x} \\ u_{2y} \\ u_{3x} \\ u_{3y} \\ u_{4x} \\ u_{4y} \end{bmatrix} \tag{4.2.4.6}$$

To better solve this problem, we can use Julia

8. Exercise 4.2.5

Find an expression for the matrix B of a three-node bar element in 2D as a function of the Jacobian and the local coordinate ξ. Consider $\varepsilon = \frac{\partial u}{\partial s}$, where s is the curvilinear coordinate along the bar path.

8.1 Solution

solution text